

CS & AI • EXPLAINED

How does Google find answers so **fast**?

It searches the whole web in 0.3 seconds — except it doesn't.

THE HOOK

You hit **enter** and in about **0.3 seconds** Google "searched the entire internet".

Except — it **didn't**. The real trick is way cleverer.

In 5 minutes, you'll see what really happens. 🔍



TODAY'S QUESTION

How does it search **billions** of
pages —
and answer before you **blink**?

It searches a pre-built **index**.

- When you search, Google does **not** go scan the live internet.
- It looks inside a giant **pre-built index**...
- ...like the **index at the back of a textbook** — which word is on which page.

PICTURE THIS

Word → list of pages it appears on.

Already built. Just look it up.

Crawlers build it **before** you search.

- Programs called **crawlers** (or “spiders”) roam the web 24/7.
- They visit pages, **follow links**, read the words...
- ...and note which word appears on which page.



visit page



follow links



read words



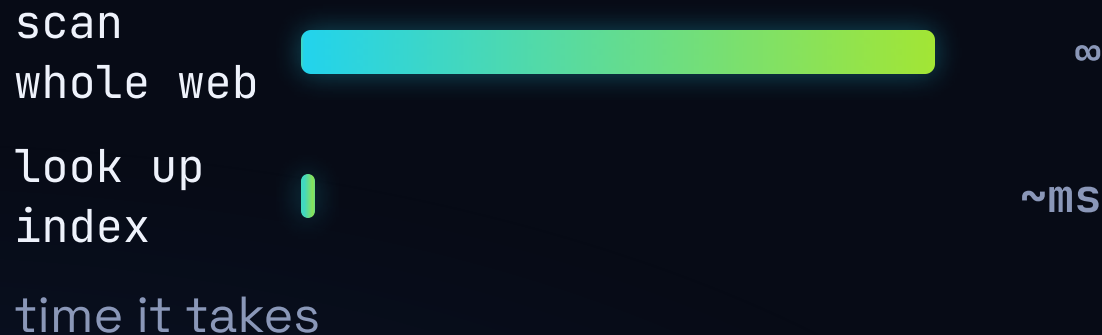
add to index

All of this happens **before** you ever type a thing.

IDEA 3 / 4 — WHY THE LOOKUP IS INSTANT

Looking up the index is **milliseconds**.

Find your words across the web:



- **Scan whole web** = impossibly slow.
- **Look up the index** = a few milliseconds.
- Just like finding a word via a book's *index* vs reading every page.

Finding is easy. **Ranking** is hard.

Millions of pages match your words. Which goes on top?

- The real magic is **ranking** — deciding the **order**.
- Signals: relevance · quality · links from other sites · freshness · your location.
- Finding pages is easy; ranking them *well* is the hard problem.

WHY IT MATTERS

It explains things you've noticed.

- **Brand-new pages take time** to show up — they must be crawled first.
- **Results differ** by person and place — location is a ranking signal.
- **SEO** = helping your page *rank*, so it lands higher in the order.

Search isn't one trick — it's crawl, index, then rank.

COMMON MYTH — "YEH GALAT SAMAJHTE HAIN"

X MYTH

"Google **scans the entire internet** the moment you search."

✓ REALITY

It searches its own **pre-built index**. The crawling and indexing already happened **in advance** — your search just looks it up.

RECAP — 2 THINGS TO REMEMBER

- 1 Google **crawls + indexes** the web ahead of time.
- 2 Your search hits the **index** (fast); **ranking** decides the order.

Crawl, index, **rank**. That's the magic. 🔍

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Next question: **What is machine learning, simply?** 🤖